The University of the South Pacific

Serving the Cook Islands, Fiji, Kiribati, Marshall Islands, Nauru, Niue, Samoa, Solomon Islands, Tokelau, Tonga, Tuvalu, and Vanuatu.

School of Information Technology, Engineering, Mathematics, and Physics

CS311: Operating Systems

Test 1 Solution – SEMESTER 1, 2022

Time Allowed: 50 mins

30 marks

Weight: 10%

INSTRUCTIONS

1. Answer all questions in this booklet. All questions are compulsory.
2. Total number of pages including this cover sheet is 5.

1) Explain the relationship between the operating system and the computer hardware. [2 marks]

The operating system helps to make computer hardware available to the application programs. Without Operating System we cannot access computer hardware.

2) Why do microkernel operating systems have a performance disadvantage over those that employ monolithic kernels? [2 Marks]

OS components within a microkernal operating system use processes that communicate via messages which incur message passing overheads. This is achieved by function calls in a monolithic kernel which is much faster.

3) Explain the purpose of the system calls? [2 Marks]

System calls allow user-level processes to request services of the operating system.

4) How does the distinction between kernel mode and user mode function as a fundamental form of protection (security) system? [2 marks]

The distinction between kernel mode and user mode provides a rudimentary form of protection in the following manner. Certain instructions could be executed only when the CPU is in kernel mode. Similarly, hardware devices could be accessed only when the program is executing in kernel mode. Control over when interrupts could be enabled or disabled is also possible only when the CPU is in kernel mode. Consequently, the CPU has very limited capability when executing in user mode, thereby enforcing protection of critical resources.

5) Explain why a modular kernel may be the best of the current operating system design techniques. [3 marks]

The modular approach combines the benefits of both the layered and microkernel design techniques. In a modular design, the kernel needs only to have the capability to perform the required functions and know how to communicate between modules. However, if more functionality is required in the kernel, then the user can dynamically load modules into the kernel. The kernel can have sections with well-defined, protected interfaces, a desirable property found in layered systems. More flexibility can be achieved by allowing the modules to communicate with one another.

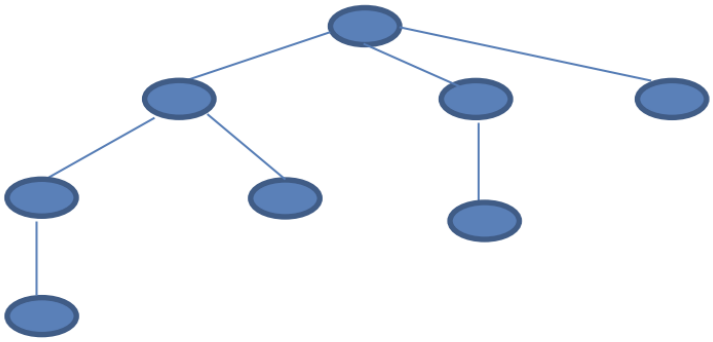
6) Multi-programming (or multi-tasking) enables more than a single process to apparently execute simultaneously. How is this achieved on a uniprocessor? [2 marks]

Preemptive multitasking is achieved by rapidly context switching between processes. Each process is given a window of time to execute, called a 'timeslice'. The processor then gets interrupted by a timer, which causes it to context switch to another process, which will run until it blocks, or until the next tick. By switching rapidly enough, it creates the illusion that several processes are running simultaneously.

7) Construct a process tree for the following. [4 marks]

```
main()
{
 int pid = fork();
 if(pid == 0)
     fork();
 else
     fork();

 fork();
}
```



8) Use the following program to answer questions that follow:

```
1. #include <pthread.h>
2. void *adding(void *);
3. int x = 0;
4. main()
5. {
6. int
7. num[] = {2, 5};
8. pthread_t thread[2];
9. pthread_create(&thread[0], NULL, adding, (void *)num[0]);
10. pthread_create(&thread[1], NULL, adding, (void
    *)num[1]);
11. pthread_exit(NULL);
12. }
13. void *adding(void *data){
14. x += (int) data;
15. printf("%d\n", x);
16. pthread_exit(NULL);
17. }
```

a. How many threads will be created during the execution of the program? [1 mark]

2 threads

b. What is the purpose for instruction at line numbers 10 and 11? [1 mark]

10: creates a new thread and executes "adding" routine with parameter 2 and 5.

11: terminates the currently running thread

c. What could be the output of the above program? [2 marks]

2 7 OR 5 7

9) Consider the following attempt to solve the critical section problem for 2 threads t0 and t1. "turn" is a global shared variable which can have values either 0 or 1.

//thread t0 uses this code
while(turn!=0);
//enter critical section
turn = 1;
	//thread t1 uses this code
	while(turn!=1);
	//enter critical section
	turn = 0;

a. Explain whether the above solution provides mutual exclusion for the critical section? [2 marks]

Yes, it does. The variable 'turn' is either T0 or T1, and if it is T0, then only thread T0 can progress past 'critical_section_enter_t0', and vice versa

b. Is this proposal a satisfactory solution to critical section problem? Justify your answer [3 marks]

No, it is not. It does not guarantee progress. In particular, thread T0 cannot reenter the critical section after it has left it unless thread T1 entered and left the critical section in the interim. Consider:

critical_section_enter_t0();
critical_section_leave_t0();

//may block even though t1 is not in the critical section
critical_section_enter_t0();
critical_section_leave_t0();

10) Determine with reasons whether the following can be used as a solution for the producer-consumer problem? [4 marks]

`Producer() {`	`Consumer() {`
`wait(emptyBuffers);`	`wait(fullBuffers);`
`wait(mutex);`	`wait(mutex);`
`place item;`	`remove item;`
`signal(fullBuffers);`	`signal(emptyBuffers);`
`signal(mutex);`	`signal(mutex);`
`}`	`}`

This code is correct. this code allows more concurrency which is the advantage to coding in this manner. Since the mutex immediately surrounds both the Producer and Consumer actions of putting an item and taking an item, from the buffer, if we release the mutex quickly, we achieve better concurrency.

THE END